

MALLA REDDY UNIVERSITY

IV Year B Tech I Semester

L/T/P/C
3/0/0/3

(MR20-1CS0158) DEEP LEARNING AND APPLICATIONS

COURSE OBJECTIVES:

1. Understand complexity of Deep Learning algorithms and their limitations.
2. Understand modern notions in data analysis oriented computing.
3. Be capable of confidently applying common Deep Learning algorithms in practice and implementing their own.
4. Be capable of performing distributed computations.
5. Be capable of performing experiments in Deep Learning using real-world data.

UNIT-I:

Machine Learning Basics: Learning Algorithms, Capacity, Over fitting, and Under fitting, Hyper parameters and Validation Sets, Estimators, Bias and Variance, Maximum Likelihood Estimation, Bayesian Statistics, Supervised and Unsupervised Learning algorithms, Stochastic Gradient Descent, Building a ML algorithm, Challenges and Motivation to Deep learning.

UNIT-II:

Tensor Flow: Computational Graph, Key highlights, Creating a Graph, Regression example, Gradient Descent, Tensor Board, Modularity, Sharing Variables, Keras Perceptrons: What is a Perceptron, XOR Gate.

Activation Functions: Sigmoid, ReLU, Hyperbolic Fns, Softmax

UNIT-III:

Artificial Neural Networks: Introduction, Perceptron Training Rule, Gradient Descent Rule.

Gradient Descent and Back propagation: Gradient Descent, Stochastic Gradient Descent, Back propagation, Some problems in ANN Optimization and Regularization: Over fitting and Capacity, Cross Validation, Feature Selection, Regularization, Hyper parameters.

UNIT-IV:

Convolutional Neural Networks: Introduction to CNNs, Kernel filter, Principles behind CNNs, Multiple Filters, CNN applications

Recurrent Neural Networks: Introduction to RNNs, Unfolded RNNs, Seq2Seq RNNs, LSTM, RNN applications.

UNIT-V:

Autoencoders: Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Deep Learning applications: Image Processing, Natural Language Processing, Speech Recognition, Video Analytics.

TEXT BOOKS:

1. Goodfellow, I., Bengio, Y., and Courville, A., Deep Learning, MIT Press, 2016.

REFERENCE BOOKS:

1. Bishop, C., M., Pattern Recognition and Machine Learning, Springer, 2006.
2. Yegnanarayana, B., Artificial Neural Networks PHI Learning Pvt. Ltd, 2009.
3. Golub, G., H., and Van Loan, C., F., Matrix Computations, JHU Press, 2013.
4. Satish Kumar, Neural Networks: A Classroom Approach, Tata McGraw-Hill Education, 2004

COURSE OUTCOMES:

1. Learn topics such as convolutional neural networks, recurrent neural networks, training deep networks and high-level interfaces.
2. Build deep learning models in Tensor Flow and interpret the results.
3. Understand the language and fundamental concepts of artificial neural networks.
4. Troubleshoot and improve deep learning models.
5. Build own deep learning project.
6. Differentiate between machine learning, deep learning and artificial intelligence.